

PORTUGUESE BANKING INSTITUTION

Banking Campaign Data Analysis

Code Transcript

Dataset: banking_data.csv | 45,211 clients | 18 features

Period: May 2008 – November 2010

Language: Python 3 | pandas · matplotlib · seaborn

Questions covered: Q1 – Q18 (Data Exploration & Correlation Analysis)

In [1]:

```
#Importing necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:

```
#Import csv file (read it)
df = pd.read_csv('/content/banking_data.csv')
```

In [3]:

```
#Lets see if the file is correct by checking row and column
df.shape
```

Out [3]:

```
(45216, 19)
```

In [4]:

```
df.head(-1)
```

Out [4]:

```
age      job  marital marital_status  education default  balance  \
0        58  management  married      married  tertiary    no    2143
1        44  technician  single      single  secondary    no      29
2        33  entrepreneur  married      married  secondary    no      2
3        47  blue-collar  married      married  unknown    no   1506
4        33      unknown  single      single  unknown    no      1
...      ...      ...      ...      ...      ...      ...      ...
45210    37  entrepreneur  married      married  secondary    no   2971
45211    29  management  single      single  tertiary    no    765
45212    68      retired  married      married  secondary    no   1146
45213    53  management  married      married  tertiary    no    583
45214    73      retired  married      married  secondary    no   2850

housing loan  contact  day month  day_month  duration  campaign  pdays  \
0      yes   no  unknown    5   may    5-May    261         1    -1
1      yes   no  unknown    5   may    5-May    151         1    -1
2      yes  yes  unknown    5   may    5-May     76         1    -1
3      yes   no  unknown    5   may    5-May     92         1    -1
4      no   no  unknown    5   may    5-May    198         1    -1
...      ...  ...      ...      ...      ...      ...      ...      ...
45210   no   no  cellular   17  nov    17-Nov    361         2   188
45211   no   no  cellular   16  nov    16-Nov    238         1    -1
45212   no   no  cellular   16  nov    16-Nov    212         1   187
45213   no   no  cellular   17  nov    17-Nov    226         1   184
45214   no   no  cellular   17  nov    17-Nov    300         1    40

previous poutcome  y
0         0  unknown  no
1         0  unknown  no
2         0  unknown  no
3         0  unknown  no
4         0  unknown  no
...      ...      ...
45210    11   other  no
45211     0  unknown  yes
45212     6  success  yes
45213     4  success  yes
45214     8  failure  yes
```

```
[45215 rows x 19 columns]
```

In [5]:

```
df.isnull().sum()
```

Out [5]:

```
age          0
job          0
marital      3
marital_status 3
education    3
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
day_month    0
duration     0
campaign     0
pdays       0
previous     0
poutcome     0
y            0
dtype: int64
```

In [6]:

```
df['marital_status']=df['marital_status'].fillna('unknown')
df['marital']= df['marital'].fillna('unknown')
df['education']=df['education'].fillna('unknown')
df.isnull().sum()
```

Out [6]:

```
age          0
job          0
marital      0
marital_status 0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
day_month    0
duration     0
campaign     0
pdays       0
previous     0
poutcome     0
y            0
dtype: int64
```

In [7]:

```
df.dtypes
```

Out [7]:

```
age          int64
job          object
marital      object
marital_status object
education    object
default      object
balance      int64
housing      object
loan         object
contact      object
day          int64
month        object
day_month    object
duration     int64
campaign     int64
pdays       int64
previous     int64
poutcome     object
y            object
dtype: object
```

In [8]:

```
df = df.drop(columns=['marital_status'])
```

In [9]:

```
df.shape
```

Out [9]:

```
(45216, 18)
```

In [10]:

```
#Q1
print("--- Q1: Age Stats ---")
print("Youngest client:", df['age'].min())
print("Oldest client:", df['age'].max())
print("Average (mean) age:", round(df['age'].mean(),1))
```

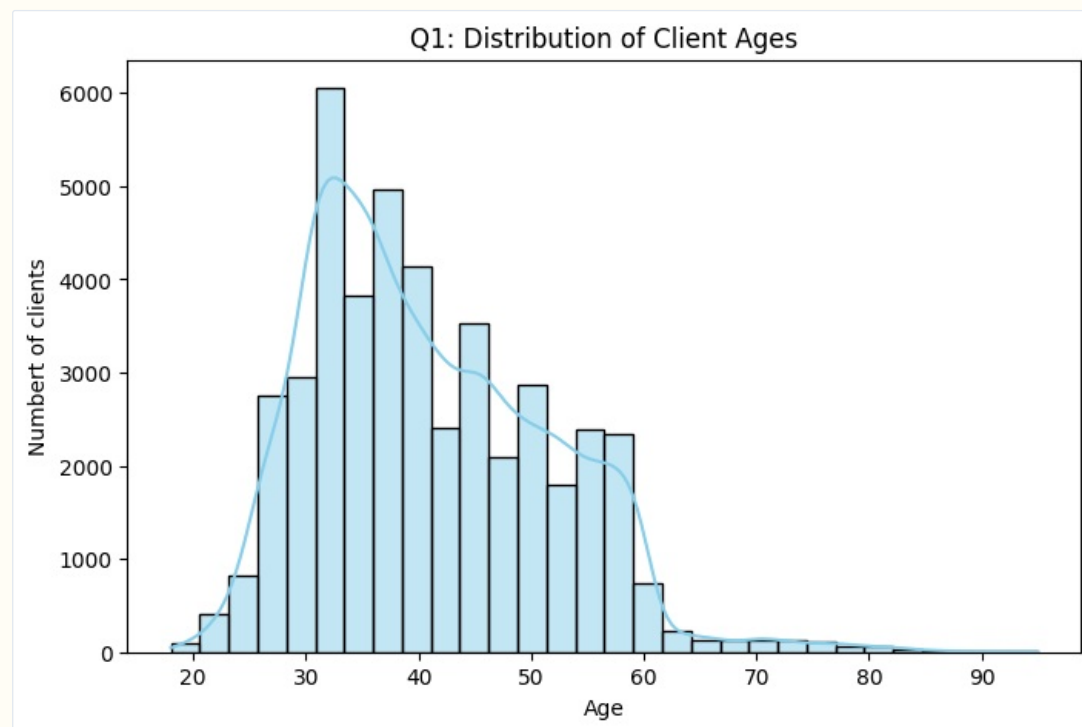
```
plt.figure(figsize=(8,5))
sns.histplot(df['age'],bins=30, kde=True,color='skyblue')
plt.title('Q1: Distribution of Client Ages')
plt.xlabel('Age')
plt.ylabel('Numbert of clients')
plt.show()
```

```
print("""\nQ1: The age distribution of the clients is slightly right-skewed. The majority of the targeted clients fall between the ages of 30 and 50, indicating that the bank is primarily focusing its marketing efforts on middle-aged, working professionals.""")
```

Out [10]:

```
--- Q1: Age Stats ---
Youngest client: 18
Oldest client: 95
Average (mean) age: 40.9
```

Out [10]: [chart]



Out [10]:

```
Q1: The age distribution of the clients is slightly right-skewed. The majority of the targeted clients fall between the ages of 30 and 50, indicating that the bank is primarily focusing its marketing efforts on middle-aged, working professionals.
```

In [11]:

```
# 1. Calculate and print the exact percentages for each job
print("--- Q2: Job Type Percentages ---")
job_percentages = df['job'].value_counts(normalize=True)*100
print(round(job_percentages,1))

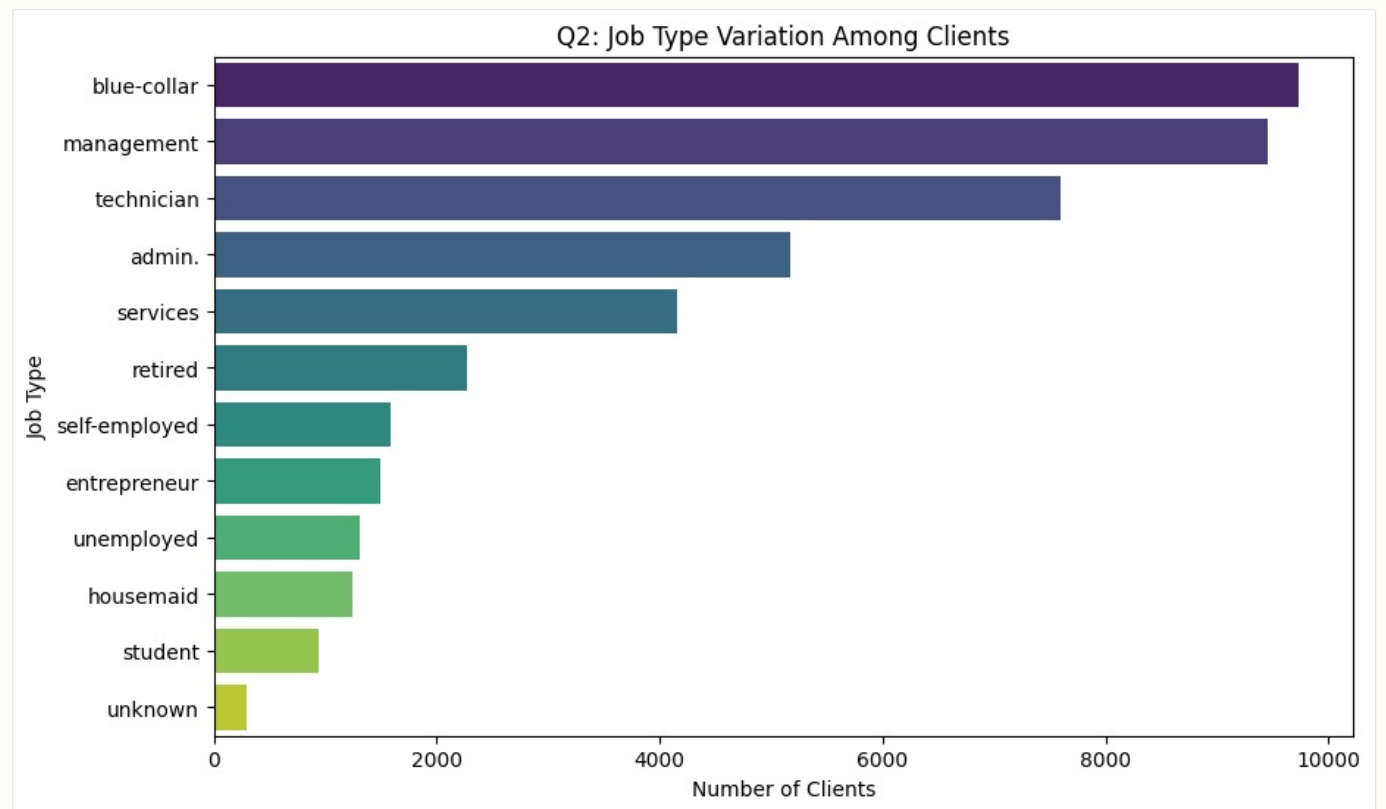
#2 plot the bar chart
plt.figure(figsize=(10,6))
sns.countplot(data=df , y='job', order=df['job'].value_counts().index, palette = 'viridis')
plt.title('Q2: Job Type Variation Among Clients')
plt.xlabel('Number of Clients')
plt.ylabel('Job Type')
plt.show()

print("""\nQ2: The clients belong to a wide variety of occupations, but the most common job types are 'blue-collar', 'management', and 'technician'. This shows the campaign is heavily reaching out to the working class and corporate professionals.""")
```

Out [11]:

```
--- Q2: Job Type Percentages ---
job
blue-collar      21.5
management       20.9
technician        16.8
admin.           11.4
services          9.2
retired           5.0
self-employed     3.5
entrepreneur      3.3
unemployed        2.9
housemaid         2.7
student           2.1
unknown           0.6
```

Out [11]: [\[chart\]](#)



Out [11]:

Q2: The clients belong to a wide variety of occupations, but the most common job types are 'blue-collar', 'management', and 'technician'. This shows the campaign is heavily reaching out to the working class and corporate professionals.

In [12]:

```
#Q3 - 'Q3: Marital Status Distribution of Clients'
#Print marital_percentage
print("---Marital Status Percentage---")
marital_percentages = df['marital'].value_counts(normalize=True)*100
print(round(marital_percentages,1))

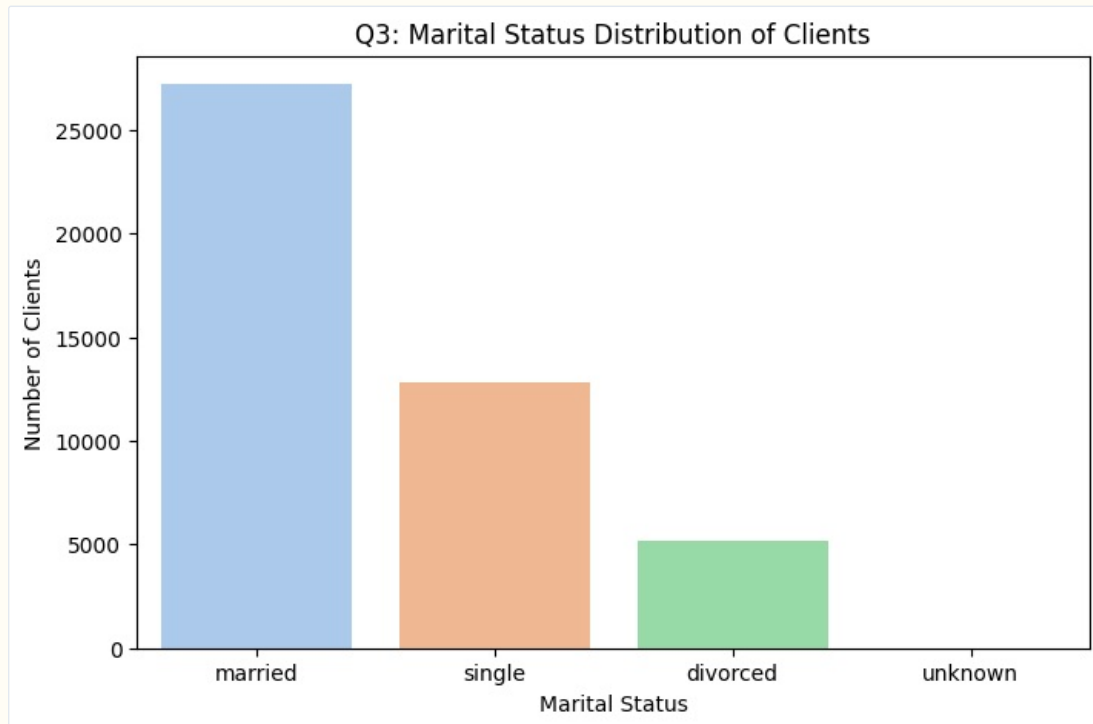
#2.Plot the Bar Chart
plt.figure(figsize=(8,5))
sns.countplot(data=df,x='marital', order=df['marital'].value_counts().index, palette='pastel')
plt.title('Q3: Marital Status Distribution of Clients')
plt.xlabel('Marital Status')
plt.ylabel('Number of Clients')
plt.show()

print("""\nQ3: The majority of the clients contacted during this campaign are married. Single and divorced clients make up a
significantly smaller portion of the dataset.""")
```

Out [12]:

```
---Marital Status Percentage---
marital
married      60.2
single       28.3
divorced     11.5
unknown      0.0
```

Out [12]: [chart]



Out [12]:

Q3: The majority of the clients contacted during this campaign are married. Single and divorced clients make up a significantly smaller portion of the dataset.

In [13]:

```
#Q4: Education Level of Clients
#Find percentage of education level of clients
print("--- Q4: Education Level Percentages ---")
education_percentage = df['education'].value_counts(normalize = True) * 100
print(round(education_percentage,1))

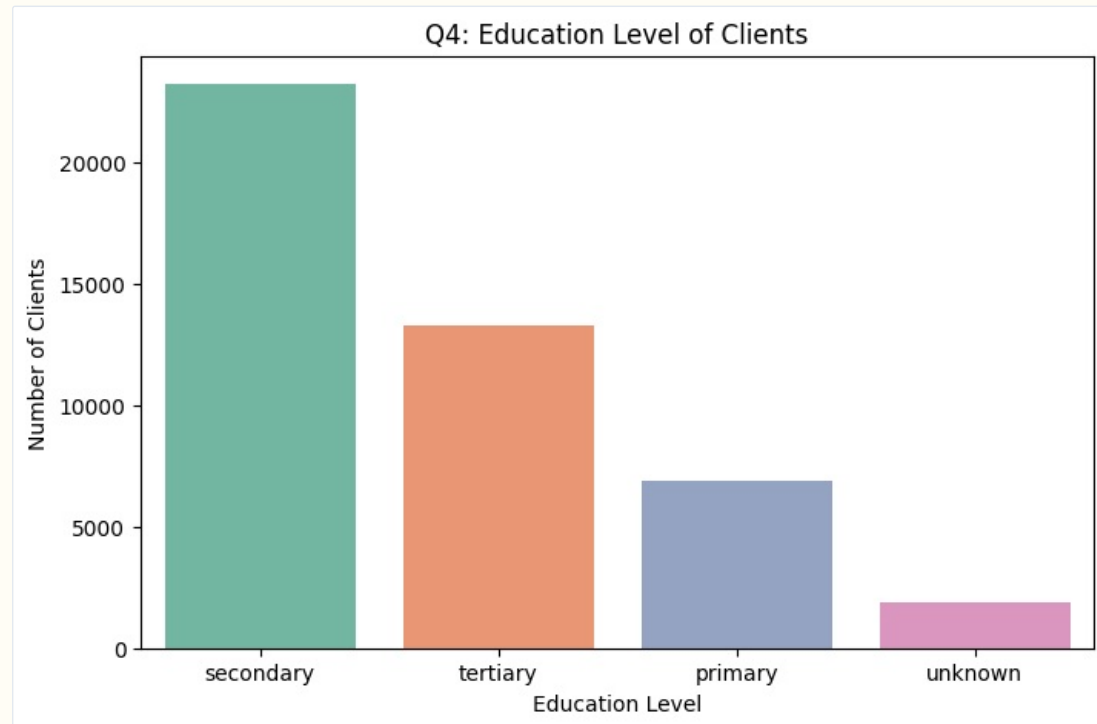
# 2. Plot the Bar Chart
plt.figure(figsize=(8,5))
sns.countplot(data=df , x='education', order=df['education'].value_counts().index , palette='Set2' )
plt.title('Q4: Education Level of Clients')
plt.xlabel('Education Level')
plt.ylabel('Number of Clients')
plt.show()

print("""\nQ4: The highest level of education for the majority of clients is 'secondary' (high school), followed closely by 'tertiary' (college/university). Very few clients have only a primary level of education.""")
```

Out [13]:

```
--- Q4: Education Level Percentages ---
education
secondary    51.3
tertiary     29.4
primary      15.2
unknown       4.1
```

Out [13]: [chart]



Out [13]:

Q4: The highest level of education for the majority of clients is 'secondary' (high school), followed closely by 'tertiary' (college/university). Very few clients have only a primary level of education.

In [14]:

```
#Q5: Proportion of Clients with Credit in Default
print("--- Q5: Credit Default Proportions ---")
default_percentages = df['default'].value_counts(normalize=True)*100
print(round(default_percentages,1))

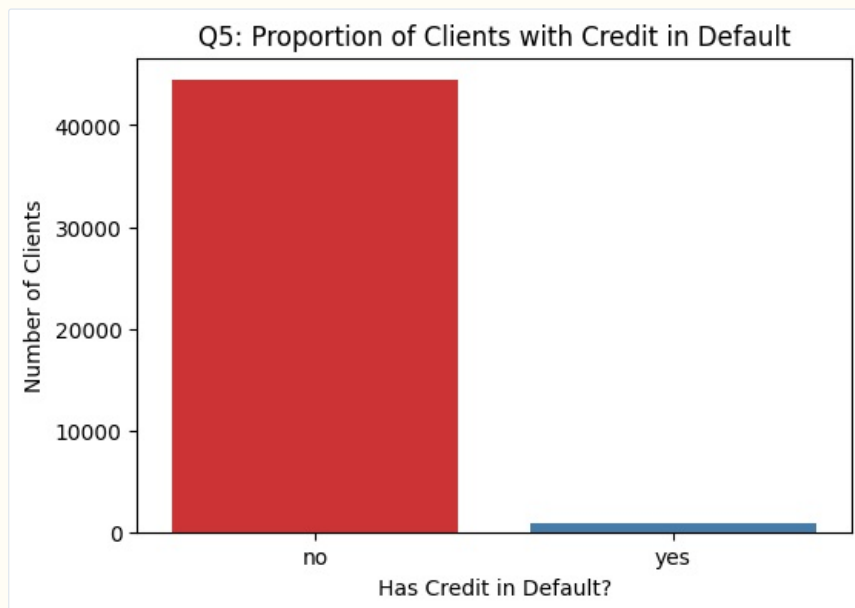
# 2. Plot the Bar Chart
plt.figure(figsize=(6,4))
sns.countplot(data=df, x='default',order = df['default'].value_counts().index,palette = 'Set1' )
plt.title('Q5: Proportion of Clients with Credit in Default')
plt.xlabel('Has Credit in Default?')
plt.ylabel('Number of Clients')
plt.show()

print("""\nQ5: The data is heavily imbalanced regarding credit defaults. An overwhelming majority of the clients do not have credit in default, which makes sense as the bank likely filters out high-risk individuals before calling them for investment opportunities.""")
```

Out [14]:

```
--- Q5: Credit Default Proportions ---
default
no      98.2
yes      1.8
```

Out [14]: [\[chart\]](#)



Out [14]:

Q5: The data is heavily imbalanced regarding credit defaults. An overwhelming majority of the clients do not have credit in default, which makes sense as the bank likely filters out high-risk individuals before calling them for investment opportunities.

In [15]:

```
print("--- Q6: Average Yearly Balance Stats ---")
print("Lowest Balance (In Debt):", df['balance'].min())
print("Highest Balance:", df['balance'].max())
print("Average(Mean)", round(df['balance'].mean(),2))
print("Median (True Middle):", df['balance'].median())

# 2. Plot the Histogram AND Boxplot side-by-side
fig,axes = plt.subplots(1,2,figsize=(14,5))

#Histogram
sns.histplot(df['balance'], bins=50, kde=True,ax=axes[0], color='lightgreen')
axes[0].set_title('Q6: Histogram of Yearly Balance')
axes[0].set_xlabel('Balance')
axes[0].set_ylabel('Number of Clients')

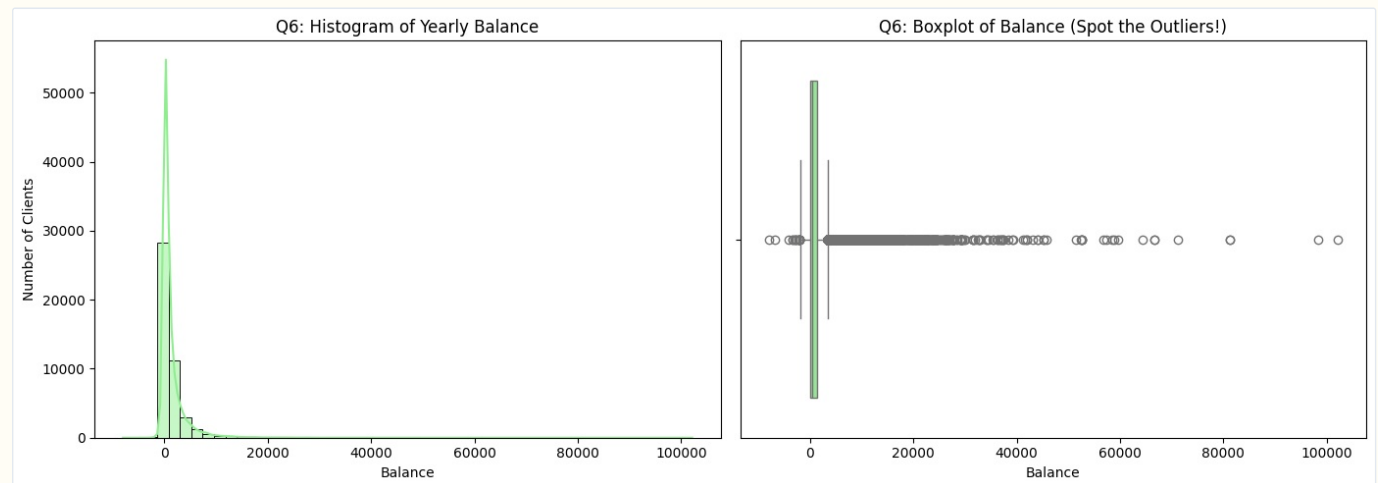
#Boxplot
sns.boxplot(x=df['balance'], ax=axes[1], color='lightgreen')
axes[1].set_title('Q6: Boxplot of Balance (Spot the Outliers!)')
axes[1].set_xlabel('Balance')

plt.tight_layout()
plt.show()
print("""\nQ6: The distribution of yearly balances is highly right-skewed. Most clients maintain relatively low bank balances, but there is a long tail of extreme outliers representing a few very wealthy individuals with massive balances.""")
```

Out [15]:

```
--- Q6: Average Yearly Balance Stats ---
Lowest Balance (In Debt): -8019
Highest Balance: 102127
Average(Mean) 1362.28
Median (True Middle): 448.5
```

Out [15]: [\[chart\]](#)



Out [15]:

Q6: The distribution of yearly balances is highly right-skewed. Most clients maintain relatively low bank balances, but there is a long tail of extreme outliers representing a few very wealthy individuals with massive balances.

In [16]:

```
# 1. Print the exact percentages for Housing Loans
print("--- Q7: Housing Loans ---")
housing_percents = df['housing'].value_counts(normalize=True)*100
print(round(housing_percents,1))

# 2. Print the exact percentages for Personal Loans
print("\n--- Q8: Personal Loans ---")
loan_percentages = df['loan'].value_counts(normalize=True) * 100
print(loan_percentages.round(1))

fig , axes = plt.subplots(1,2, figsize=(12,5))

#Q7 Chart
sns.countplot(data = df, x='housing', order=df['housing'].value_counts().index, ax=axes[0], palette='Blues')
axes[0].set_title('Q7: Clients with Housing Loans')
axes[0].set_xlabel('Has Housing Loan?')
axes[0].set_ylabel('Number of Clients')

# Q8 Chart
sns.countplot(data = df, x= 'loan', order=df['loan'].value_counts().index, ax=axes[1], palette="Oranges")
axes[1].set_title('Q8: Clients with Personal Loans')
axes[1].set_xlabel('Has Personal Loan?')
axes[1].set_ylabel('Number of Clients')

plt.tight_layout()
plt.show()

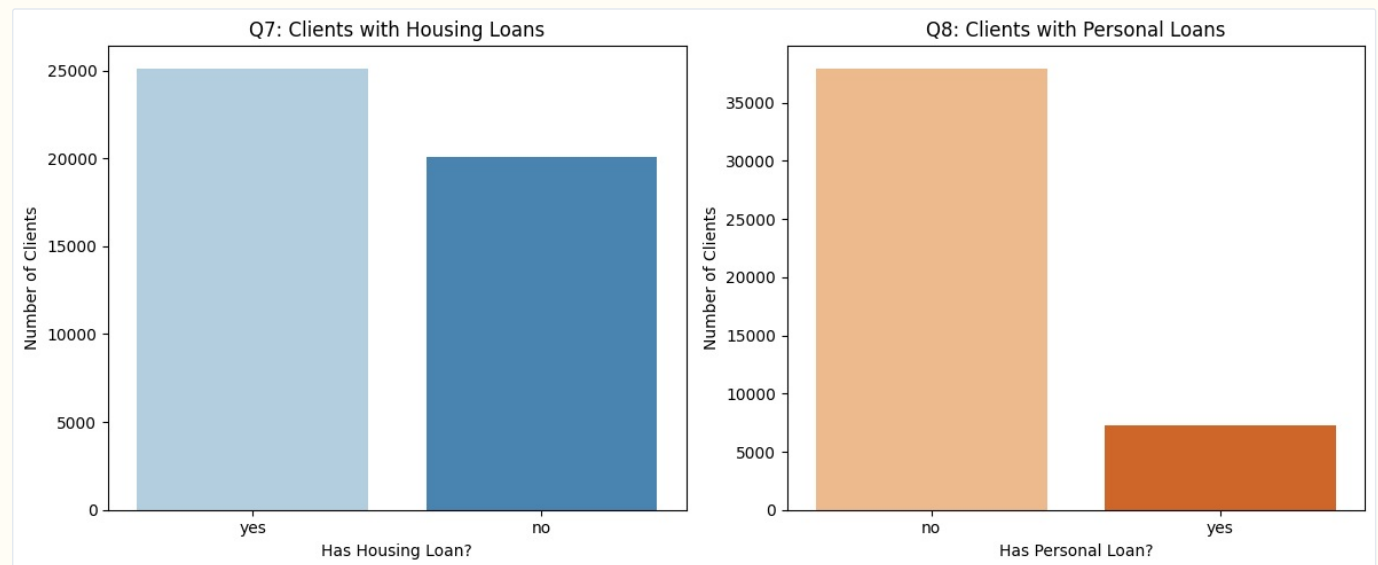
print("""\nQ7: A slight majority of the clients contacted currently have an active housing loan (mortgage). This indicates that the
bank's target audience largely consists of homeowners.""")
print("""\nQ8: Unlike housing loans, the vast majority of clients do not have an active personal loan.""")
```

Out [16]:

```
--- Q7: Housing Loans ---
housing
yes      55.6
no       44.4

--- Q8: Personal Loans ---
loan
no      84.0
yes     16.0
```

Out [16]: [\[chart\]](#)



Out [16]:

Q7: A slight majority of the clients contacted currently have an active housing loan (mortgage). This indicates that the bank's target audience largely consists of homeowners.

Q8: Unlike housing loans, the vast majority of clients do not have an active personal loan.

In [17]:

```
#Q9: Communication Types Used During Campaign

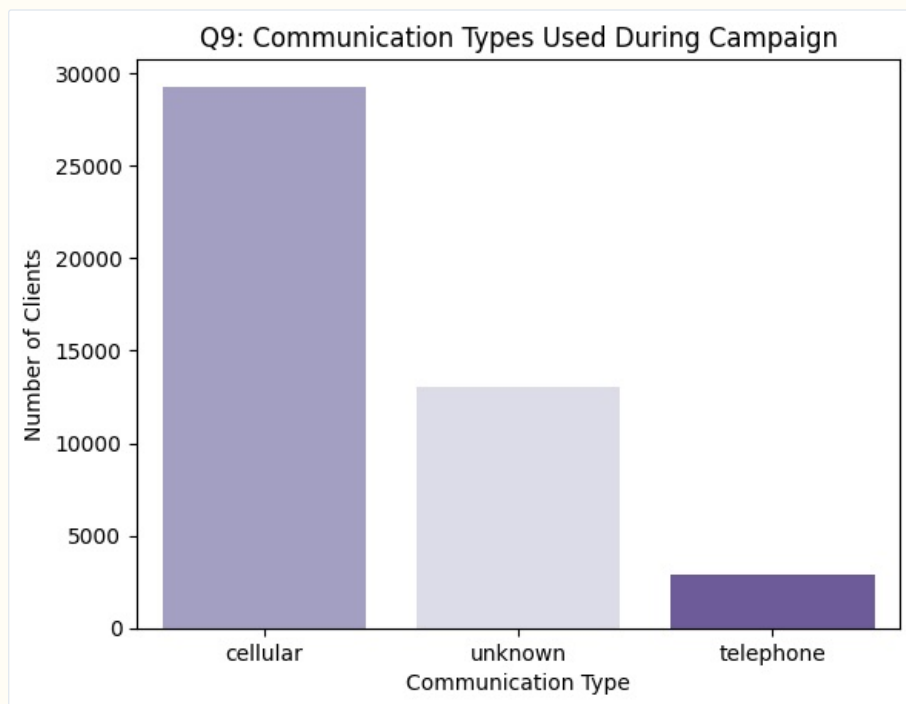
print("--- Q9: Communication Types ---")
comm_percentages = df['contact'].value_counts(normalize=True)*100
print(round(comm_percentages,1))

#Plot bar chart
sns.countplot(data=df, x='contact', hue='contact', order=df['contact'].value_counts().index, palette='Purples', legend=False)
plt.title('Q9: Communication Types Used During Campaign')
plt.xlabel('Communication Type')
plt.ylabel('Number of Clients')
plt.show()
print("""\nThe preferred and most successful method of communication by far is 'cellular' (mobile phones). Reaching clients via
traditional landlines ('telephone') is much less common.""")
```

Out [17]:

```
--- Q9: Communication Types ---
contact
cellular      64.8
unknown       28.8
telephone      6.4
```

Out [17]: *[chart]*



Out [17]:

The preferred and most successful method of communication by far is 'cellular' (mobile phones). Reaching clients via traditional landlines ('telephone') is much less common.

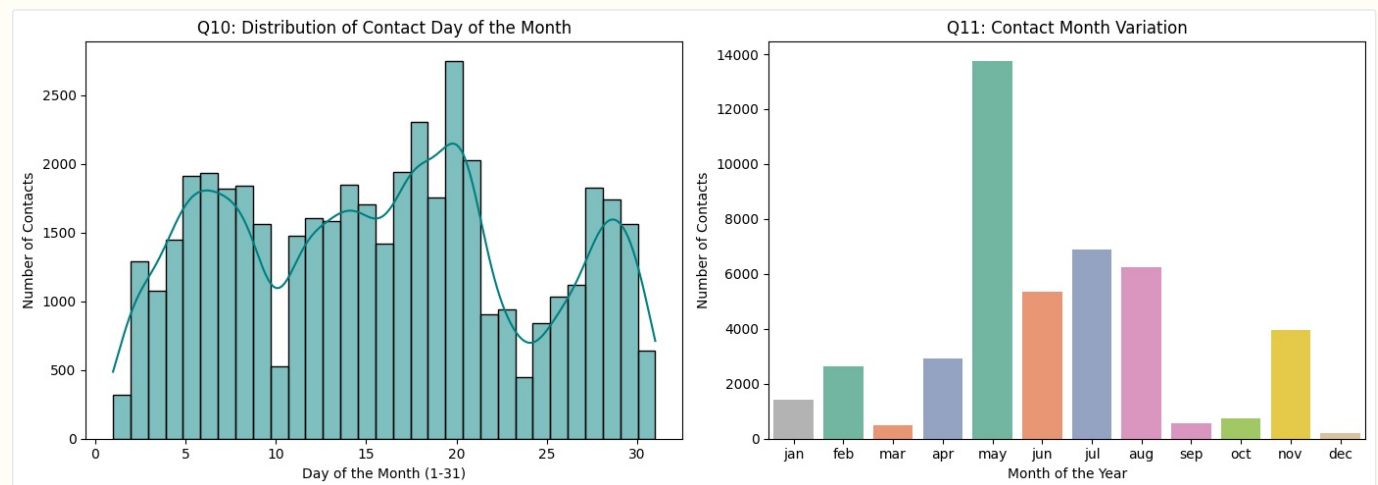
In [18]:

```
#Q10 Chart: Histogram for the Day (Numerical)
fig,axes = plt.subplots(1,2,figsize=(14,5))
sns.histplot(data=df ,x='day', bins = 31,kde=True, ax=axes[0],color='teal')
axes[0].set_title('Q10: Distribution of Contact Day of the Month')
axes[0].set_xlabel('Day of the Month (1-31)')
axes[0].set_ylabel('Number of Contacts')

# Q11 Chart: Bar Chart for the Month (Categorical)
month_order = ['jan', 'feb', 'mar', 'apr', 'may', 'jun', 'jul', 'aug', 'sep', 'oct', 'nov', 'dec']
sns.countplot(data=df, x='month', hue='month', order=month_order,ax=axes[1],palette='Set2', legend=False)
axes[1].set_title('Q11: Contact Month Variation')
axes[1].set_xlabel('Month of the Year')
axes[1].set_ylabel('Number of Contacts')

plt.tight_layout()
plt.show()
print("""\nQ10: The contacts are spread out across all days of the month, but there are noticeable spikes in campaign activity during
the middle of the month (around the 15th to the 20th).""")
print("""\nQ11: 'May' is by far the busiest month for this marketing campaign, seeing more than double the activity of most other
months. Conversely, the winter months like December and January saw very little campaign activity.""")
```

Out [18]: [chart]



Out [18]:

Q10: The contacts are spread out across all days of the month, but there are noticeable spikes in campaign activity during the middle of the month (around the 15th to the 20th).

Q11: 'May' is by far the busiest month for this marketing campaign, seeing more than double the activity of most other months. Conversely, the winter months like December and January saw very little campaign activity.

In [19]:

```
df['pdays'].head(-1)
```

Out [19]:

```
0      -1
1      -1
2      -1
3      -1
4      -1
...
45210  188
45211  -1
45212  187
45213  184
45214   40
Name: pdays, Length: 45215, dtype: int64
```

In [20]:

```
#Q12: Distribution of Last Contact Duration
# 1. Print the exact stats
print("--- Q12: Call Duration Stats ---")
print("Shortest Call:", df['duration'].min(), "seconds")
print("Longest Call:", df['duration'].max(), "seconds")
print("Average (Mean) Call:", round(df['duration'].mean(), 1), "seconds")
print("Median Call:", df['duration'].median(), "seconds")

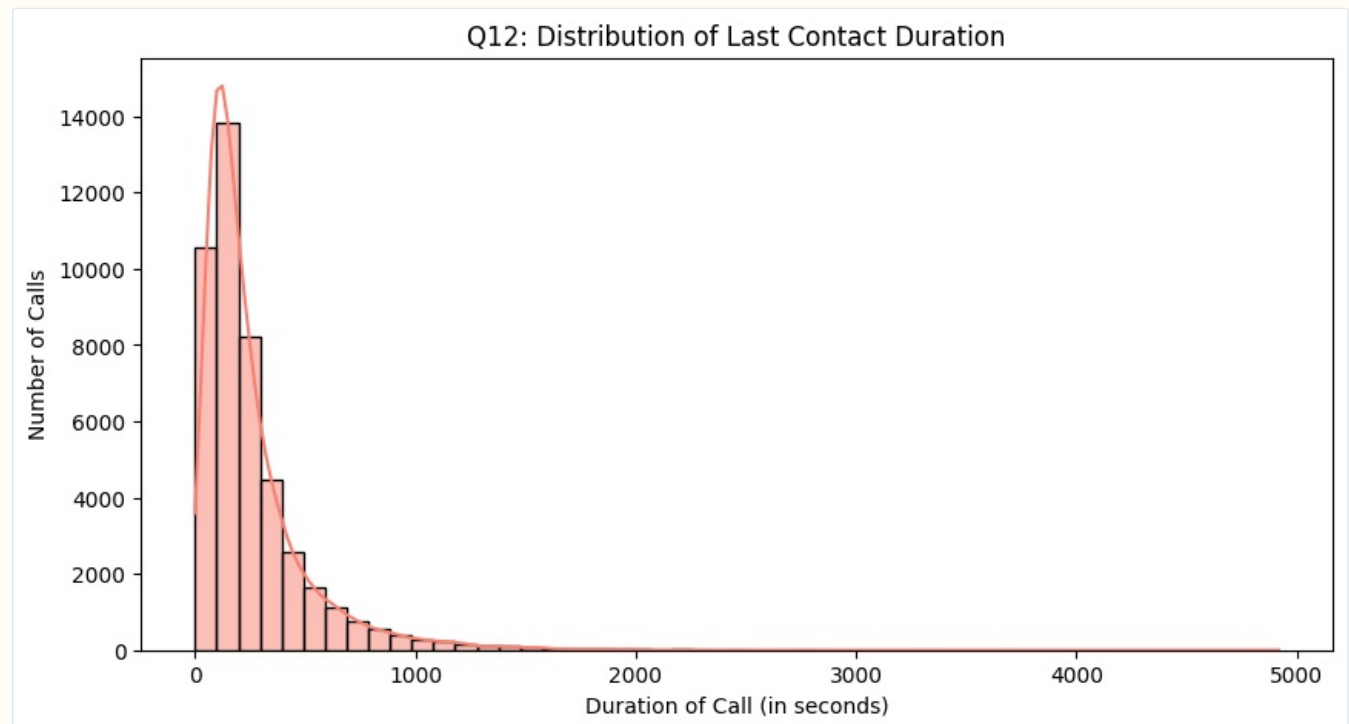
#Plot The Histogram
plt.figure(figsize=(10,5))
sns.histplot(data=df, x='duration', bins=50, kde=True, color='salmon')
plt.title('Q12: Distribution of Last Contact Duration')
plt.xlabel('Duration of Call (in seconds)')
plt.ylabel('Number of Calls')
plt.show()

print("\nThe distribution of the last contact duration is highly right-skewed. The vast majority of phone calls are relatively short, with the median call lasting only 180 seconds (3 minutes). However, there is a long tail indicating a small number of extremely long conversations, with the maximum call duration reaching nearly 4,918 seconds (over 1 hour and 20 minutes!).")
```

Out [20]:

```
--- Q12: Call Duration Stats ---
Shortest Call: 0 seconds
Longest Call: 4918 seconds
Average (Mean) Call: 258.2 seconds
Median Call: 180.0 seconds
```

Out [20]: [\[chart\]](#)



Out [20]:

The distribution of the last contact duration is highly right-skewed. The vast majority of phone calls are relatively short, with the median call lasting only 180 seconds (3 minutes). However, there is a long tail indicating a small number of extremely long conversations, with the maximum call duration reaching nearly 4,918 seconds (over 1 hour and 20 minutes!).

In [21]:

```
#Q13- How many contacts were performed during the campaign for each client?
print("--Q13 Contacts performed during campaign stats--")
print("Lowest Number", df['campaign'].min())
print("Highest Number", df['campaign'].max())
print("Mean Number", df['campaign'].mean().round(2))
print("Median Number", df['campaign'].median())

print("\nHistogram")
plt.figure(figsize=(12,5))
sns.histplot(data=df,x='campaign',bins=30,kde=True,color='lightgreen')
plt.title('Q13: Number of Contacts Performed During Campaign')
plt.xlabel('Number of Contacts')
plt.ylabel('Number of Clients')
plt.show()

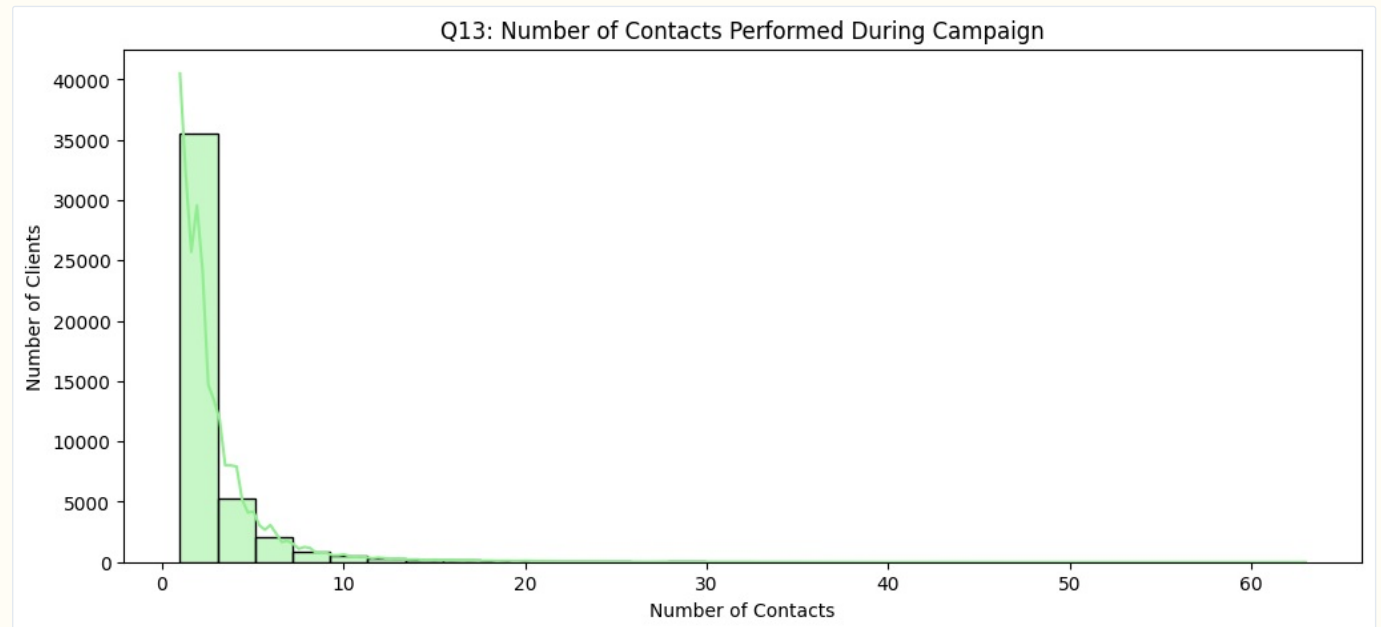
print("\nMost clients were only contacted 1, 2, or 3 times during this specific campaign. However, the distribution is heavily right-skewed due to a few extreme outliers where some clients were called aggressively (upwards of 50+ times).")
```

Out [21]:

```
--Q13 Contacts performed during campaign stats--
Lowest Number 1
Highest Number 63
Mean Number 2.76
Median Number 2.0
```

Histogram

Out [21]: [chart]



Out [21]:

Most clients were only contacted 1, 2, or 3 times during this specific campaign. However, the distribution is heavily right-skewed due to a few extreme outliers where some clients were called aggressively (upwards of 50+ times).

In [22]:

```
#Q14-- What is the distribution of the number of days passed since the client was last contacted from a previous campaign?
```

```
print("===Client was last contacted from a previous campaign stats===")
print("Lowest pdays", df['pdays'].min())
print("Highest pdays", df['pdays'].max())
print("Mean pdays",df['pdays'].mean().round(2))
print("Median pdays",df['pdays'].median())
```

```
print("\nHistogram")
# 2. Histogram
plt.figure(figsize=(12, 5))
sns.histplot(data=df, x='pdays', bins=50, kde=True, color='skyblue')
plt.title('Q14: Days Passed Since Client Was Last Contacted')
plt.xlabel('Number of Days (pdays)')
plt.ylabel('Number of Clients')
plt.show()
```

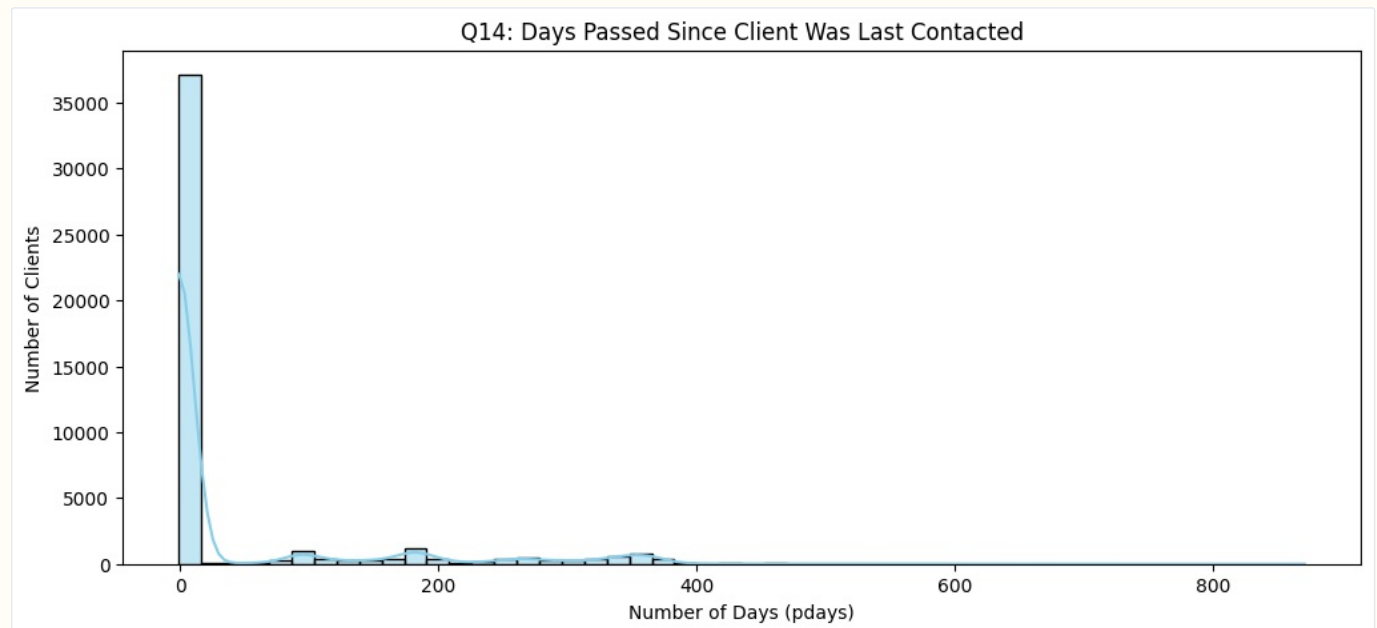
```
print("The pdays variable shows how many days passed since a client was contacted in a previous campaign. The histogram reveals a massive, overwhelming spike at -1. In this dataset, -1 means the client was never previously contacted. Because the vast majority of people in this campaign are new contacts, the median is exactly -1. For the small minority of clients who were contacted before, the days passed usually range from 100 to 400 days.")
```

Out [22]:

```
===Client was last contacted from a previous campaign stats===
Lowest pdays -1
Highest pdays 871
Mean pdays 40.2
Median pdays -1.0
```

Histogram

Out [22]: [\[chart\]](#)



Out [22]:

The pdays variable shows how many days passed since a client was contacted in a previous campaign. The histogram reveals a massive, overwhelming spike at -1. In this dataset, -1 means the client was never previously contacted. Because the vast majority of people in this campaign are new contacts, the median is exactly -1. For the small minority of clients who were contacted before, the days passed usually range from 100 to 400 days.

In [23]:

```
#Q15-- - How many contacts were performed before the current campaign for each client?
print("=== Q15: Contacts Performed Before Current Campaign Stats ===")
print("Lowest previous contacts:", df['previous'].min())
print("Highest previous contacts:", df['previous'].max())
print("Mean previous contacts:", df['previous'].mean().round(2))
print("Median previous contacts:", df['previous'].median())

print("\nHistogram")
# 2. Histogram
plt.figure(figsize=(12, 5))
sns.histplot(data=df, x='previous', bins=50, kde=True, color='purple')
plt.title('Q15: Number of Contacts Before Current Campaign')
plt.xlabel('Number of Previous Contacts')
plt.ylabel('Number of Clients')
plt.show()

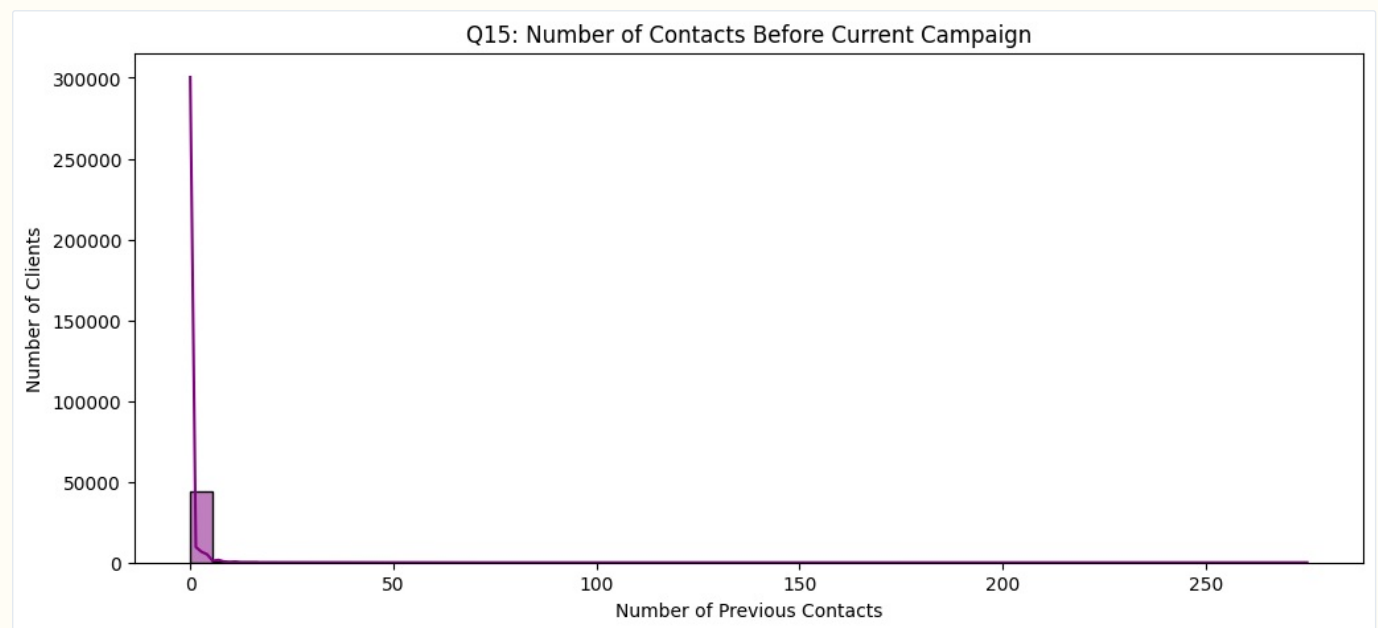
print("""\nHow many contacts were performed before the current campaign for each client?
Write-up: Similar to pdays, this distribution is heavily right-skewed with a massive spike at 0.
This confirms again that the overwhelming majority of clients had exactly zero previous contacts
prior to this current campaign.""")
```

Out [23]:

```
=== Q15: Contacts Performed Before Current Campaign Stats ===
Lowest previous contacts: 0
Highest previous contacts: 275
Mean previous contacts: 0.58
Median previous contacts: 0.0
```

Histogram

Out [23]: *[chart]*



Out [23]:

How many contacts were performed before the current campaign for each client?
Write-up: Similar to pdays, this distribution is heavily right-skewed with a massive spike at 0.
This confirms again that the overwhelming majority of clients had exactly zero previous contacts
prior to this current campaign.

In [24]:

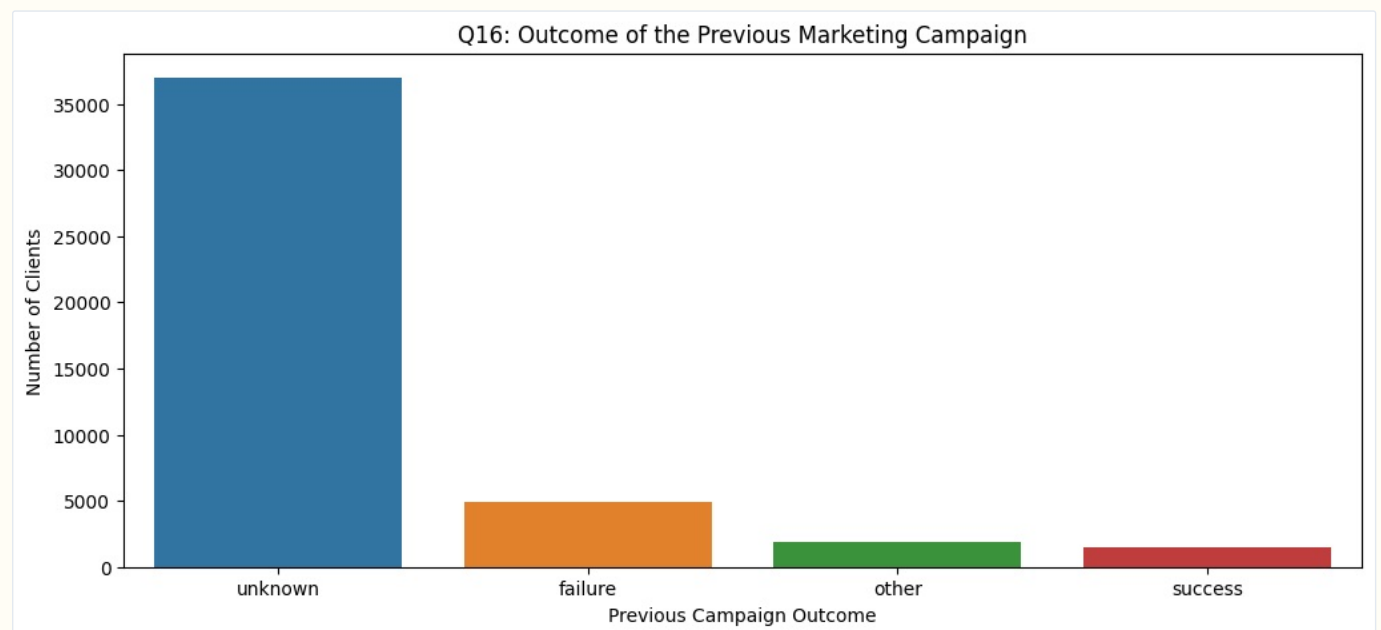
```
print("=== Q16: Previous Campaign Outcome ===")
percentage_poutcome = df['poutcome'].value_counts(normalize=True)*100
print(round(percentage_poutcome,1))
plt.figure(figsize=(12,5))
sns.countplot(data=df , x='poutcome',hue='poutcome', order=df['poutcome'].value_counts().index)
plt.title('Q16: Outcome of the Previous Marketing Campaign')
plt.xlabel('Previous Campaign Outcome')
plt.ylabel('Number of Clients')
plt.show()

print("The overwhelming majority of previous campaign outcomes are categorized as 'unknown' (over 80%). This perfectly aligns with our findings from the previous questions, which revealed that most of these clients are brand new leads who were never contacted prior to the current campaign. For the small fraction of clients who were contacted previously, the most common known outcome was 'failure', with a much smaller percentage resulting in a 'success'.")
```

Out [24]:

```
=== Q16: Previous Campaign Outcome ===
poutcome
unknown    81.7
failure    10.8
other       4.1
success     3.3
```

Out [24]: *[chart]*



Out [24]:

The overwhelming majority of previous campaign outcomes are categorized as 'unknown' (over 80%). This perfectly aligns with our findings from the previous questions, which revealed that most of these clients are brand new leads who were never contacted prior to the current campaign. For the small fraction of clients who were contacted previously, the most common known outcome was 'failure', with a much smaller percentage resulting in a 'success'.

In [25]:

```
print("=== Q17: Term Deposit Subscription (Target Variable) ===")
percentage_y = df['y'].value_counts(normalize=True) * 100
print(percentage_y.round(1))

plt.figure(figsize=(12,5))
sns.countplot(data=df, x='y', hue='y', order=df['y'].value_counts().index)
plt.title('Q17: Did the Client Subscribe to a Term Deposit?')
plt.xlabel('Subscribed (Yes / No)')
plt.ylabel('Number of Clients')

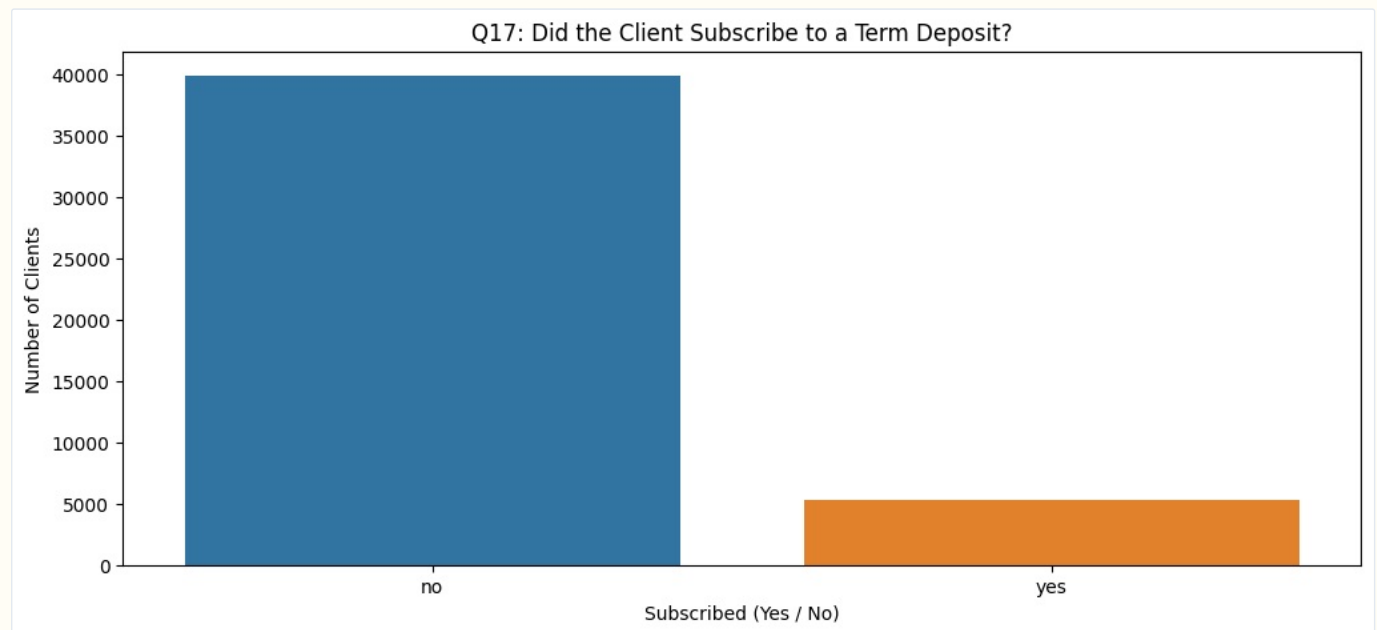
plt.show()

print("The y variable represents the ultimate success of the campaign: whether the client subscribed to a term deposit. The data reveals that the campaign has a relatively low success rate, which is typical for telemarketing. The dataset is highly imbalanced, with the vast majority of clients (approx. 88.3%) declining the offer ('no'), while only a small fraction (approx. 11.7%) actually subscribed ('yes').")
```

Out [25]:

```
=== Q17: Term Deposit Subscription (Target Variable) ===
y
no      88.3
yes     11.7
```

Out [25]: [\[chart\]](#)



Out [25]:

The y variable represents the ultimate success of the campaign: whether the client subscribed to a term deposit. The data reveals that the campaign has a relatively low success rate, which is typical for telemarketing. The dataset is highly imbalanced, with the vast majority of clients (approx. 88.3%) declining the offer ('no'), while only a small fraction (approx. 11.7%) actually subscribed ('yes').

In [26]:

```
df['y_numeric'] = df['y'].map({'yes':1, 'no':0})

numeric_df = df.select_dtypes(include=['int64', 'float64'])

corr_matrix = numeric_df.corr()

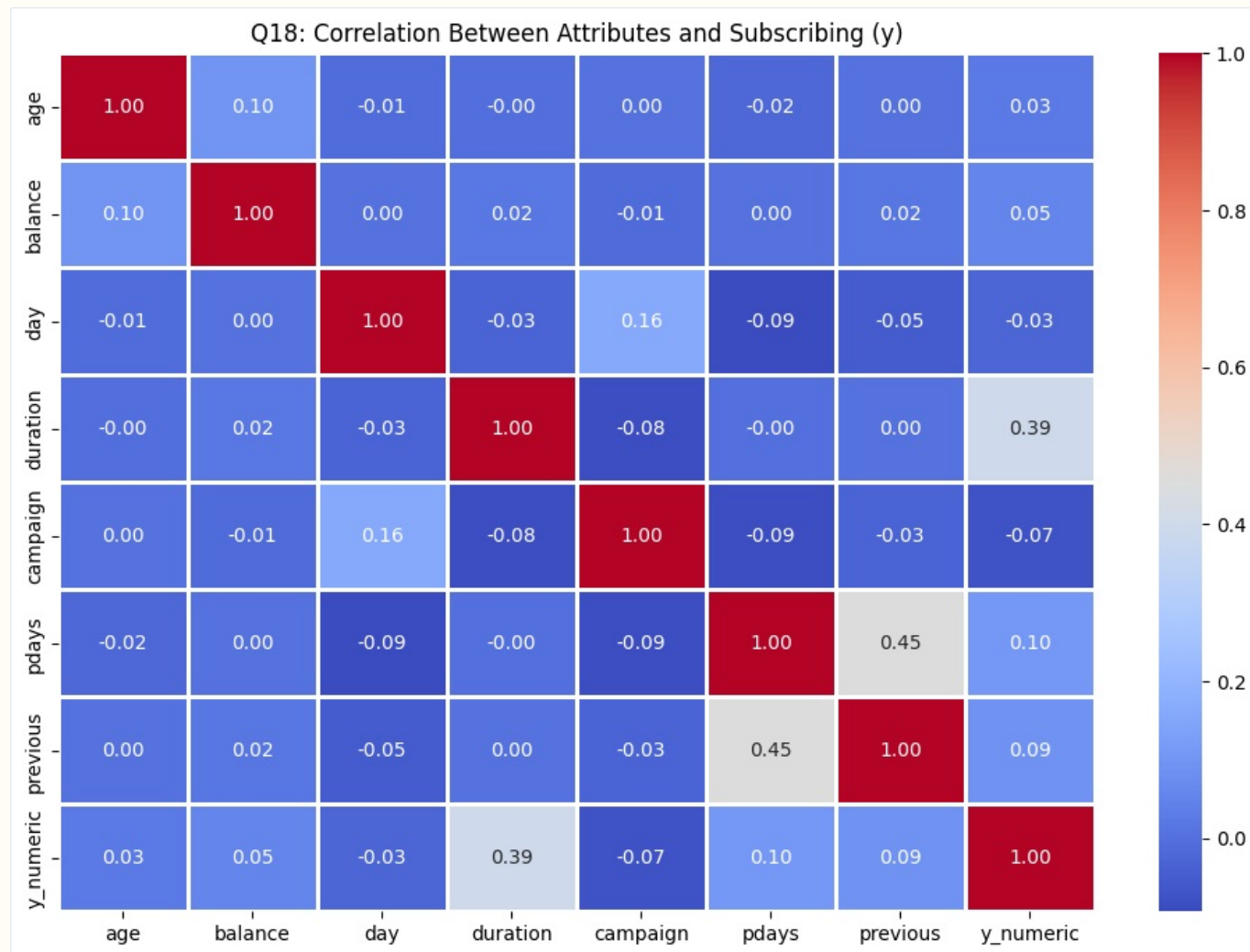
plt.figure(figsize=(12, 8))
# annot=True puts the numbers in the boxes, cmap='coolwarm' makes positives red and negatives blue
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap='coolwarm', linewidths=1)
plt.title('Q18: Correlation Between Attributes and Subscribing (y)')
plt.show()

print("\n--- Correlation with Target Variable (Subscribing) ---")
print(corr_matrix['y_numeric'].sort_values(ascending=False))

print("""\n\nTo determine the correlations between different attributes and the likelihood of a client subscribing, we converted the target variable (y) into a numeric format (1 for yes, 0 for no) and generated a correlation heatmap. The analysis reveals that most numerical attributes—such as age, balance, and campaign (number of calls)—have virtually no correlation with the success of the campaign, scoring very close to 0.

However, there is one significant exception: duration. The duration of the last contact has the strongest positive correlation with a successful subscription. This indicates a clear business insight: the longer a client remains on the phone during the marketing call, the significantly higher the likelihood that they will subscribe to the term deposit.""")
```

Out [26]: [chart]



Out [26]:

```
--- Correlation with Target Variable (Subscribing) ---
y_numeric    1.000000
duration     0.394387
pdays       0.103699
previous     0.093576
balance      0.052821
age          0.025648
day         -0.028307
campaign     -0.073294
Name: y_numeric, dtype: float64
```

To determine the correlations between different attributes and the likelihood of a client subscribing, we converted the target variable (y) into a numeric format (1 for yes, 0 for no) and generated a correlation heatmap. The analysis reveals that most numerical attributes—such as age, balance, and campaign (number of calls)—have virtually no correlation with the success of the campaign, scoring very close to 0. However, there is one significant exception: duration. The duration of the last contact has the strongest positive correlation with a successful subscription. This indicates a clear business insight: the longer a client remains on the phone during the marketing call, the significantly higher the likelihood that they will subscribe to the term deposit.